

Java to Python Translation via Semantic IR and Pseudocode

Java → Python Translation Using Semantic Intermediate Representation

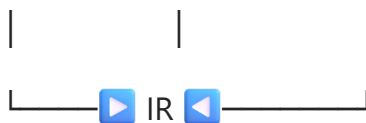
This document describes a **robust, research-aligned process** for converting Java source code to Python using a structured pseudocode / semantic intermediate representation (IR). The intended audience is a Python developer with general familiarity with ASTs and code generation.

1. Why Use Pseudocode or an Intermediate Representation?

Direct Java-to-Python translation is fragile because Java and Python differ heavily in typing, control flow expression, and object models. Instead of comparing Java code directly to Python code, this approach compares *intent*.

The strategy is:

Java code —▶ IR —▶ Python code



By normalizing both Java and Python into the same IR, you verify that they express identical semantics rather than identical syntax.

This approach is supported by compiler design principles and modern translation-validation research. Studies on pseudocode-based translation show improved correctness when an intermediate semantic layer is used instead of direct translation.

2. Key Design Principle

Do not generate informal, English-like pseudocode.

Instead, generate a **restricted, structured pseudocode** that behaves like a compiler IR but reads clearly to humans.

This avoids ambiguity while still allowing inspection and diffing.

3. Overall Pipeline

1. Parse Java source into a Java AST
 2. Normalize Java AST into a language-neutral IR
 3. Generate Python code from the IR
 4. Parse generated Python into a Python AST
 5. Normalize Python AST into the same IR
 6. Compare IR trees to validate correctness
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4. Recommended Tools

Java Parsing

- **javalang** – Pure Python Java parser targeting Java 8; easy to walk and analyze ASTs
- **ANTLR (Java grammar)** – Heavier setup but supports newer Java versions and full grammar control

Python Parsing

- `ast` (standard library) – Parse Python into ASTs, supports unparse from Python 3.9+

Formatting / Output

- **black** – Format generated Python code for readability
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5. Designing the Intermediate Representation

The IR should be:

- Language-neutral
- Deterministic
- Explicit about control flow and data flow

This aligns with standard IR categories used in source-to-source compilers: structural and linear representations.

Example IR Node Types

Function(name, parameters, body)

Block(statements)

Assign(target, expression)

If(condition, then_block, else_block)

While(condition, body)

Call(name, arguments)

Return(value)

Notably absent:

- Java-specific features (types, visibility, annotations)
 - Python-specific features (`self`, decorators)
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6. Structured Pseudocode Generation

Pseudocode is generated **from the IR**, not directly from source code.

Example

```
function sum_list(items):  
  
    total = 0  
  
    index = 0  
  
    while index < length(items):  
  
        total = total + items[index]  
  
        index = index + 1  
  
    return total
```

This same representation can be derived from both Java and Python, enabling direct semantic comparison.

7. Validation Strategy

Primary Check

- Normalize both Java and Python into IR
- Compare IR trees for structural equivalence

What This Catches Well

- Missing or extra branches
- Incorrect loop conditions
- Dropped return statements
- Changed mutation order

What It Does Not Catch

- Performance differences
- Library/API mismatches
- Threading or concurrency behavior

This tradeoff is consistent with research on IR-based translation validation systems.

8. Suggested Initial Scope

For an initial implementation, support:

- Functions and methods
- Conditionals
- Loops (normalize all to `while`)
- Assignments and expressions
- Returns

Defer handling:

- Exceptions
 - Generics
 - Lambdas
 - Inheritance hierarchies
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9. Summary

Using a structured pseudocode / semantic IR as a verification layer:

- Aligns with compiler architecture
- Improves translation correctness
- Makes validation explicit and inspectable

The key is discipline: treat pseudocode as a **machine-checkable semantic model**, not prose.